Library Models and Pricing

Andrea Carapelli

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Scope

The market handles, the simulation models and the products of the library, as implemented: definitions, pricing formulas and algorithms, calibration and the treatment of the greeks. Implementation (classes, market data layout) is in the companion document *tQuant — Architecture*; questions and exercises in *tQuant — Exercises*.

1 Market handles

The market data the pricers and the models read: curves, volatilities, the equity forward. Every quote is an input of the prices, and the sensitivities of a price are its derivatives w.r.t. these quotes.

1.1 Interest-rate curves

A curve holds continuously compounded zero rates r_i on pillars T_i (Act/365), linear in between and flat outside the pillars (definitions of Brigo and Mercurio, 2006, ch. 1):

$$P(0,T) = e^{-r(T)\,T}, \qquad r(T) = r_i + \frac{r_{i+1} - r_i}{T_{i+1} - T_i}\,(T - T_i) \quad (T_i \le T < T_{i+1}).$$

- zero rate $r(T) = -\ln P(0,T)/T$, with $r(0) = r_1$;
- simple forward $F(0;T_1,T_2) = \big(P(0,T_1)/P(0,T_2) - 1\big)/\tau$, $\tau = T_2 - T_1$ in the curve day count;
- instantaneous forward $f(0,T) = -\partial_T \ln P(0,T) = r(T) + T\,r'(T)$: at a pillar the right derivative, $f = r_1$ before the first pillar and r_n after the last.

Each currency has an overnight curve (ESTR, SONIA, SOFR), the discount curve of everything in that currency; an Ibor index has its own projection curve (EUR 6M) for its forward fixings.

Bootstrap. The instruments are built from the quotes (deposits and OIS for the overnight curve; FRAs and swaps for EUR 6M), with one pillar at the end date of each. The pillar rates are solved by Newton's method until every instrument has zero NPV, each priced as in §3.1; the Jacobian $\partial \mathrm{NPV}_i/\partial r_j$ of the final step links the pillar sensitivities to the quote sensitivities.

1.2 Interest-rate volatilities

- *Caplet vols*: normal (or lognormal) vols on fixing expiry × absolute strike nodes, bilinear, flat outside.
- *Swaption cube*: vols on option expiry × swap tenor × strike offset $K - F$ from the forward swap rate, trilinear, flat outside.
- *Caplet stripping*: from flat cap vols (one vol for all the caplets of a cap) to caplet vols, cap by cap in increasing maturity, each new caplet vol solving the price of the next cap; the inverse gives the flat vols of a caplet surface.

1.3 Equity spot, forward and dividends

An equity has a spot S_0, a repo q_r, either cash dividends D_i with ex-dates t_i and payment dates p_i or a continuous yield q, and the discount curve $P(0,\cdot)$ of its currency. The forward, the same for every pricer and model:

$$\text{cash dividends:}\quad F(T) = \frac{\big(S_0 - \mathrm{PV}_{\mathrm{div}}(T)\big)\,e^{-q_r T}}{P(0,T)}, \qquad \mathrm{PV}_{\mathrm{div}}(T) = \sum_{t_i \le T} D_i\,P(0,p_i);$$

$$\text{yield:}\quad F(T) = \frac{S_0\,e^{-(q+q_r)T}}{P(0,T)}.$$

A dividend is in the horizon by its ex-date (the stock drops then) and discounted from its payment date (the ex-date if none is given). The equivalent dividend yield to T is $q_{\mathrm{eff}}(T) =$

$-\ln\left(1-\mathrm{PV}_{\mathrm{div}}(T)/S_0\right)/T$. Cash dividends are used when the schedule is given (the yield is then ignored), the yield otherwise.

1.4 Equity volatilities

Black implied vols σ_{ij} on maturities T_i (years) $\times$ absolute strikes K_j. For the closed-form pricers the vol at (K, T) is bilinear in (K, T) and flat outside the grid, so it depends on the four surrounding nodes. The local vol model builds its own representation from the same nodes, in total variance and forward log-moneyness (§2.3).

2 Models

The simulation models. Each one is described by its dynamics, how it is calibrated to the market, its closed-form prices when it has them, the simulation scheme and how its sensitivities are defined.

Common conventions. The equity models are written on the *forward ratio*

$$X_t = \frac{S_t}{F(t)}, \qquad X_0 = 1, \qquad S_t = F(t)\, X_t,$$

with $F(t)$ the market forward of §1.3. Under the risk-neutral measure X is a martingale, so every model reproduces the market forward at every date by construction: rates, repo and dividends enter only through $F(t)$, and a cash dividend is a drop of S proportional to its level. In Monte Carlo the simulated X_t can optionally be rescaled to sample mean one at each date, which removes the noise on the forward.

Sensitivities. A sensitivity is the derivative of the price w.r.t. one market quote, all the other quotes held, *with the model recalibrated*: it is the limit of "bump the quote, recalibrate, reprice with the same random numbers". The quotes are the spot, the repo, the cash dividend amounts, the zero rates at the curve pillars and the implied vol at each node (T_i, K_j) of the surface. Holding the other quotes fixes the scenario convention: when the spot moves, the implied vols stay at their *absolute* strikes (sticky strike), the dividends stay as cash amounts and the curves as zero rates (another convention, e.g. vols fixed in moneyness, gives another delta). Through a calibration by optimisation the derivative of the parameters comes from the implicit function theorem at the optimum; through an explicit construction (local vol) from the chain rule. In Monte Carlo the derivative is taken path by path (pathwise estimator, Giles and Glasserman, 2006; Glasserman, 2003), all quotes at once by algorithmic differentiation (Savine, 2018); it equals the bump derivative of the same estimator, and needs a payoff continuous in the path (indicators are smoothed). Units: delta per unit of spot, vega per vol point and per node, rho per basis point and per pillar.

2.1 Geometric Brownian motion

Dynamics. $\mathrm{d}S_t = \mu S_t\, \mathrm{d}t + \sigma S_t\, \mathrm{d}W_t$, with constant drift and vol: the Black–Scholes model (Black and Scholes, 1973; Merton, 1973). $\ln S_T$ is Gaussian, so a European option has the Black–Scholes price (§3.4).

Calibration. One model per option (K, T), nothing to optimise:

1. σ is the implied vol of the surface at (K, T) (bilinear between the four surrounding nodes);
2. $r = -\ln P(0,T)/T$, the zero rate to the expiry;
3. with cash dividends the simulated process is the escrowed spot $S^* = S_0 - \mathrm{PV}_{\mathrm{div}}(T)$ with drift $\mu = r - q_r$; with a dividend yield it is the spot with $\mu = r - q - q_r$.

The model reprices the option it is built for and nothing else: it is the Monte Carlo counterpart of the Black–Scholes formula, not a model of the surface. With cash dividends the escrowed path is the spot only at T, so the model is not fit for products observed before the expiry.

Simulation. Exact: $S_{t+\Delta} = S_t \exp\left((\mu - \frac{1}{2}\sigma^2)\Delta + \sigma\sqrt{\Delta}\, Z\right)$, $Z \sim N(0,1)$; one step is enough for a European payoff.

Sensitivities. The chain is short,

$$\text{quotes} \;\to\; \big(S^*,\ \mu,\ r,\ \sigma(K,T)\big) \;\to\; S_T \;\to\; \text{payoff},$$

and the simulation is exact, so the Monte Carlo sensitivities are the Black–Scholes ones up to the Monte Carlo error:

- *Delta*: $\partial S^*/\partial S_0 = 1$ and the drift does not depend on the spot, so the delta of a call is $e^{-q_r T}N(d_1)$, the Black–Scholes delta of the discrete-dividend model ($e^{-(q+q_r)T}N(d_1)$ with a yield); the vol stays at (K,T) (sticky strike).
- *Vega*: σ is the bilinear interpolation of the four nodes around (K,T), so the vega splits on these four nodes with the interpolation weights, and sums to the Black–Scholes vega.
- *Rates*: r enters the drift and the discounting through $P(0,T)$ only (the pillars around T), and the present value of the dividends through their payment dates.
- *Dividends*: $\partial S^*/\partial D_i = -P(0,p_i)$ for every ex-date before T, so $\partial V/\partial D_i = -\Delta\, P(0,p_i)$.

2.2 Displaced diffusion

Dynamics. A mixture of lognormal and normal dynamics (Rubinstein, 1983):

$$\mathrm{d}X_t = \sigma(t)\,\big(\beta(t)\,X_t + 1 - \beta(t)\big)\,\mathrm{d}W_t,$$

with β and σ piecewise constant between the surface maturities. The local vol of X is $\sigma(t)\,(\beta + (1-\beta)/X)$: $\beta = 1$ is lognormal (flat smile), $\beta \to 0$ normal, with a vol that rises as X falls (negative skew). The skew per unit log-moneyness ranges from 0 to about $-\sigma/2$, so the model cannot produce a steeper equity skew, and it has no smile curvature. With constant parameters $Y = \beta X + 1 - \beta$ is lognormal with vol $\beta\sigma$, which gives calls on X in closed form:

$$\mathbb{E}\big[(X_T - k)^+\big] = \frac{\mathrm{Black}(1,\ \beta k + 1 - \beta,\ \beta\sigma,\ T)}{\beta} \qquad (1 - k \text{ if } \beta k + 1 - \beta \le 0),$$

puts by parity.

Calibration. Two stages:

1. *Slices*: for each maturity T_i, the constant parameters $\theta_i = (\beta_{T_i}, \sigma_{T_i})$ that best fit the smile of T_i: least squares on the market calls per unit forward c_{ij} at the strike nodes $k_{ij} = K_j/F(T_i)$,

$$\min_{\theta_i}\ L_i = \tfrac{1}{2}\sum_j \big(c^{\mathrm{DD}}(k_{ij}, T_i;\ \theta_i) - c_{ij}\big)^2, \qquad \beta \in [0.02, 1.5];$$

   two parameters fit the level and the skew of the slice, not its curvature;
2. *Forward pieces*: the time-dependent parameters on $[T_{i-1}, T_i)$ that reproduce the slices. The variance adds up, $\sigma_i^2(T_i - T_{i-1}) = \sigma_{T_i}^2 T_i - \sigma_{T_{i-1}}^2 T_{i-1}$; the effective β of a maturity is an average of the $\beta(t)$ before it, weighted towards the later times (Piterbarg, 2005; Andersen and Piterbarg, 2010),

$$\beta_T = \frac{\sum_j \beta_j W_j}{\sum_j W_j}, \qquad W_j = \int_{\text{piece } j} \sigma^2(t)\, v(t)\ \mathrm{d}t, \quad v(t) = \int_0^t \sigma^2(s)\ \mathrm{d}s,$$

   solved maturity by maturity for the β_j (a linear equation each). Slices whose skews are not consistent in time give a forward β out of its range.

Simulation. Exact on each piece (shifted lognormal step); a step across pieces uses the variance-weighted β and the summed variance.

Sensitivities. The chain goes through an optimisation:

$$\text{quotes} \;\to\; (F(T_i),\; k_{ij},\; c_{ij}) \;\to\; \theta_i = (\beta_{T_i}, \sigma_{T_i}) \;\to\; (\beta_j, \sigma_j) \;\to\; X_t \;\to\; S_t = F(t)\, X_t \;\to\; \text{payoff}.$$

- *Through the slice fit* (implicit function theorem, Henrard, 2011). At the optimum the gradient of L_i is zero, $\nabla_\theta L_i(\theta_i^*, m) = 0$, for every value of the market inputs m; differentiating this identity,

$$\frac{\partial \theta_i^*}{\partial m} = -H_i^{-1}\, \frac{\partial^2 L_i}{\partial \theta\, \partial m}, \qquad H_i = \nabla_\theta^2 L_i(\theta_i^*, m),$$

  the derivative of a re-optimisation without re-optimising (a parameter on a bound stays fixed). The forward pieces are explicit functions of the slices (chain rule).
- *Vega per node*: a node (T_i, K_j) changes only c_{ij}, hence only the slice θ_i (each slice is fitted alone), then the forward pieces from T_i on. A vanilla at T_i gets vega on all the strikes of its slice, in proportion to how much each node pulls the two parameters, not only on the node at its strike as in local vol.
- *Delta*: a higher spot raises the forwards, so the paths scale and, with the vols held at absolute strikes, the nodes move in moneyness k_{ij} and the slices are refitted. A vanilla is repriced only up to the fit residual, so its delta is the Black–Scholes sticky-strike one up to the slope of that residual.
- *Rates, repo, dividends*: through the forwards, as the delta without the move of the spot.

2.3 Local volatility

Dynamics. The volatility is a deterministic function of time and spot (Dupire, 1994; Derman and Kani, 1994):

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\, \mathrm{d}t + \sigma_{\mathrm{loc}}(t, S_t)\, \mathrm{d}W_t \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}X_t}{X_t} = \sigma_{\mathrm{loc}}(t, y_t)\, \mathrm{d}W_t, \quad y_t = \ln X_t.$$

There is one source of randomness; the instantaneous vol of the stock is known once the spot is known. The function σ_{loc} is not chosen: it is *implied* by the market, as the only diffusion of this form whose distribution of S_T at every maturity is the one of the market option prices. Below, $k = K/F(T)$ and $c(k,T) = \mathbb{E}[(X_T - k)^+]$ is the undiscounted call per unit of forward.

From option prices to local vol. Three steps.

(A) *Prices give the distributions* (Breeden and Litzenberger, 1978). Differentiating $c = \int_k^\infty (x-k)\, p_T(x)\, \mathrm{d}x$ in the strike,

$$\partial_k c = -\mathbb{Q}(X_T > k), \qquad \partial_{kk} c = p_T(k)\,:$$

a butterfly $c(k-h) - 2c(k) + c(k+h)$ divided by h^2 pays a unit-area triangle around k and tends to the density. The calls of one maturity are the risk-neutral distribution at that maturity; no butterfly arbitrage $\Leftrightarrow$ c convex in k $\Leftrightarrow$ $p_T \geq 0$.

(B) *The distributions give the local vol* (Dupire, 1994). The distributions alone do not fix how the stock moves between maturities; within the diffusions above there is exactly one that has them all. Itô's formula on the payoff $(X_T - k)^+$ (Tanaka: $f' = \mathbf{1}\{x > k\}$, $f'' = \delta(x-k)$)

and expectation: the linear part has zero mean (X is a martingale) and only the kink at the strike contributes,

$$\partial_T c = \tfrac{1}{2}\,\sigma^2_{\text{loc}}(T,k)\,k^2\,p_T(k) = \tfrac{1}{2}\,\sigma^2_{\text{loc}}(T,k)\,k^2\,\partial_{kk}c \qquad \Longrightarrow \qquad \sigma^2_{\text{loc}}(T,k) = \frac{\partial_T c}{\tfrac{1}{2}\,k^2\,\partial_{kk}c}.$$

It is the Black–Scholes theta–gamma relation read in strike: the calls grow with the maturity only through the paths at the strike, in proportion to the local variance and to the density there. Known prices determine the local variance: calibration is a computation, not an optimisation.

(C) *Meaning* (Gyöngy, 1986; Derman and Kani, 1994). For any model with a stochastic instantaneous vol σ_t that reprices the same calls, $\sigma^2_{\text{loc}}(T,k) = \mathbb{E}[\sigma^2_T \mid X_T = k]$: the local variance is the expected instantaneous variance on the paths through (T,k).

The formula in implied total variance. With forward log-moneyness $y = \ln k$ and total implied variance $w(y,T) = \sigma^2_{\text{imp}}(y,T)\,T$, the market call is the Black–Scholes price $c = c_{\text{BS}}(y, w(y,T))$, which depends on y and w only. In y the denominator of Dupire is $\frac{1}{2}(\partial_{yy}c - \partial_y c)$. By the chain rule the numerator is $\partial_T c = \partial_w c_{\text{BS}}\,\partial_T w$, and the denominator, with the Black–Scholes identities, is $\partial_w c_{\text{BS}}\;g$; the common factor cancels (Gatheral, 2006, eq. 1.10):

$$\sigma^2_{\text{loc}} = \frac{\partial_T w}{g}, \qquad g = 1 - \frac{y}{w}\,\partial_y w + \frac{1}{4}\Big(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\Big)(\partial_y w)^2 + \frac{1}{2}\,\partial_{yy} w.$$

- *Numerator* (calendar term): the variance the market adds per unit of time at fixed moneyness, the forward variance.
- *Denominator* (butterfly term): g is the density relative to the lognormal one with the same total variance, $p(y) = g(y)\,e^{-d_-^2/2}/\sqrt{2\pi w}$ with $d_- = -y/\sqrt{w} - \sqrt{w}/2$. The skew term $-\frac{y}{w}\,\partial_y w$ is below 0 on the downside of a negative skew: lower g, higher local vol there.
- *No static arbitrage*, also the condition for a local vol to exist: $\partial_T w \geq 0$ at fixed y (calendar spreads) and $g > 0$ (butterflies).
- *No skew*: $g = 1$ and $\sigma^2_{\text{loc}} = \partial_T w$; a flat surface gives a flat local vol.

Calibration. There is no parameter to fit: calibrating means building, from the implied vol nodes σ_{ij} (maturities T_i, strikes K_j), a surface $w(y,T)$ defined everywhere that passes through the nodes, has continuous $\partial_y w$, $\partial_{yy} w$ in closed form, keeps $\partial_T w \geq 0$ and has wings without arbitrage; then applying the formula. The fit is exact at the nodes by construction. The steps:

1. *Coordinates.* Each node goes to $y_{ij} = \ln(K_j/F(T_i))$ and $w_{ij} = \sigma^2_{ij} T_i$ (the same strike has a different y at each maturity, since the forward moves).
2. *Smile per maturity.* $w(\cdot, T_i)$ is a natural cubic spline in y through the nodes (the least-bending curve with continuous curvature), so $\partial_y w$ and $\partial_{yy} w$ are analytic: no finite differences of option prices, which are tiny at short maturities and in the wings, and no slope jumps (linear interpolation in vol puts a spike in the density at every node). Outside the nodes it continues linearly with the slope capped at 1.9, below the bound 2 of Lee (2004) on the wings of the total variance.
3. *Between maturities.* At fixed y, w is linear in T (the same weights for $\partial_y w$, $\partial_{yy} w$), so $\partial_T w$ is the forward variance of each interval and is positive whenever the slices do not cross; before the first maturity w is linear from 0 (flat implied vol), after the last the implied vol is flat. The local vol therefore jumps at each quoted maturity.

4. *The formula on a grid.* σ_{loc} on a grid of time step 1/96 year (plus the maturities) and 300 points of $y \in [\ln 0.05, \ln 5]$, with g floored at 10^{-3} and σ_{loc} capped at 200%; the simulation reads it by bilinear interpolation at $y = \ln(S_t/F(t))$.
5. *Arbitrage check.* The cells within the quoted strikes where $g < 10^{-3}$ or $\partial_T w \leq 0$ are reported as static arbitrage: there no local vol can reprice the surface.

Between the nodes this representation (spline in total variance) is not the bilinear interpolation in vol used by the closed-form pricers of §1.4: the two agree at the nodes only.

How implied and local vol relate. They are different objects: the implied vol is the constant vol that reprices one option, the local vol the instantaneous vol at (t, S). The implied variance of (K, T) is approximately the time average of the local variance over the paths that end at the strike (Gatheral, 2006, eqs. 3.8–3.11), concentrated around the straight line from the spot to the strike in log-moneyness:

$$\sigma^2_{\text{imp}}(K,T) \approx \frac{1}{T}\int_0^T \sigma^2_{\text{loc}}(t, \tfrac{t}{T}\, y_K)\ \mathrm{d}t.$$

Consequences:

- *Rule of 2* (Derman et al., 1996): with $\sigma_{\text{loc}} \approx a + b\,y$ near the money and a short maturity, $\sigma_{\text{imp}}(y) \approx \frac{1}{y}\int_0^y (a + bz)\ \mathrm{d}z = a + \frac{b}{2}\,y$: the implied skew is about half the local skew. At longer maturities the average also runs over the term structure and the ratio moves away from 2.
- A steep short-dated implied wing requires a much higher local vol there; with a convex smile the local vol at the money is below the implied vol (the paths also visit the higher-vol wings).
- The fit is checked by repricing: options simulated under the local vol, inverted with Black, give back the market implied vols.
- The local skew decays with the maturity, so the *forward* smile of the model (the smile of S_{T_2}/S_{T_1}) is flatter than today's smile of the same length (Gatheral, 2006, ch. 3 and 8); see also Derman et al. (2016).

Simulation. Log-Euler on $\ln X$ with the local vol at the start of each step:

$$\ln X_{t+\Delta} = \ln X_t - \tfrac{1}{2}\,\sigma^2\Delta + \sigma\sqrt{\Delta}\,Z, \qquad \sigma = \sigma_{\text{loc}}(t, \ln X_t),$$

σ_{loc} bilinear on its grid. The scheme has a bias of first order in the step: on an equity surface, at the maturities from 6M to 3Y, the repriced smile is about 0.2 vol points high on average with 50 steps a year (at most 0.5), and the bias is gone with 200 steps a year; with $2 \cdot 10^4$ paths the Monte Carlo error is of a few tenths of a vol point.

Sensitivities. The local vol is rebuilt from the quotes for every price, so each sensitivity runs through the construction above (chain rule), then through the paths:

$$\begin{aligned}\text{quotes} \ \to\ & (F(T_i),\ y_{ij},\ w_{ij}) \ \to\ (w,\ \partial_y w,\ \partial_{yy} w,\ \partial_T w)(y,T)\\ \to\ & \sigma_{\text{loc}}(t,y) \ \to\ X_t \ \to\ S_t = F(t)\,X_t \ \to\ \text{payoff}.\end{aligned}$$

- *Delta.* A higher spot raises every forward ($\partial F/\partial S_0 = e^{-q_r T}/P(0,T)$ with cash dividends), so the paths scale with it; with the implied vols held at their absolute strikes, every node moves to a lower log-moneyness $y_{ij} = \ln(K_j/F(T_i))$, so the local vol seen by a path at a given spot changes too (the smile is re-read relative to the new forward). For a vanilla

the two effects add up to a simple result: the model reprices every vanilla, so its price as a function of the spot is the Black–Scholes price at the fixed market vol $\sigma(K,T)$, and the delta is the Black–Scholes sticky-strike delta (with the same dividend model). For a path-dependent product the delta also contains the shift of the local vol along the paths, which a single Black–Scholes vol cannot represent.

- *Vega per node.* A bump of the node (T_i, K_j) raises w at T_i around y_{ij} (through the spline, with small ripples on the neighbouring strikes) and changes $\partial_T w$ on the two adjacent time intervals: the local vol rises before T_i and falls after it, around that log-moneyness. A vanilla on the node has the market vega on that node and almost nothing elsewhere (its price is the Black–Scholes price at that vol); a path-dependent product collects vega on all the nodes whose region (t, y) its paths cross, with signs that can differ by maturity.
- *Rates, repo, dividends.* They enter through the forwards (and the discounting of the payoff): the same chain as the delta, without the move of the spot.

For vanillas these sensitivities match the closed-form Black–Scholes sticky-strike ones up to the Monte Carlo and discretisation errors, which is the check of the construction; the information the model adds is in the exotics.

Limits. All from the assumption of a deterministic vol:

- *Flat forward smile*: products on future smiles (forward-start options, cliquets, barriers and digitals observed at future dates in autocallables) are mispriced.
- *Smile dynamics*: after a spot move the model's smile moves opposite to the observed one (Hagan et al., 2002), so the model's own hedge is poor; in practice the surface is recalibrated and the greeks follow a market convention (sticky strike).
- *No vol factor*: no vol of vol and a perfect spot–vol dependence; models that reprice the same vanillas (Heston, stochastic local vol) give different exotic prices, which local vol cannot measure.
- *Interpolation-dependent*: between the nodes, in the wings and beyond the quoted maturities the local vol is a choice of construction; exotics that live there inherit it.
- *Noise*: the second derivative in strike amplifies quote errors and arbitrages; the local vol surface and its vegas are unstable from one calibration to the next, hence arbitrage-free parametric smiles in practice (SVI, Gatheral, 2006).
- *Short-dated skew*: largely jump risk in the market; a diffusion reproduces it only with extreme local vols in the wings.

2.4 Heston

Dynamics. The variance is a mean-reverting square-root process correlated with the spot (Heston, 1993); see Oosterlee and Grzelak (2019, ch. 8):

$$\frac{\mathrm{d}X_t}{X_t} = \sqrt{v_t}\,\mathrm{d}W_t^1, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^2, \qquad \mathrm{d}\langle W^1, W^2\rangle_t = \rho\,\mathrm{d}t,$$

five constant parameters: initial variance v_0 (level of the short maturities), long-run variance θ (level of the long ones), mean reversion κ (how fast the short-term features fade; half-life $\ln 2/\kappa$), vol of vol ξ (curvature, and size of the skew), correlation ρ (sign and slope of the skew). The expected variance is $\mathbb{E}[v_t] = \theta + (v_0 - \theta)e^{-\kappa t}$. The variance stays positive when $2\kappa\theta \geq \xi^2$ (Feller); otherwise it touches zero and bounces back, which equity calibrations usually imply.

Implied vol structure.

- *ATM term structure.* With $\rho = 0$, given the path of v the stock is lognormal with variance $\int_0^T v_t \, \mathrm{d}t$, so the call is the Black–Scholes price averaged over the average variance $\bar{v}_T$ (Hull and White, 1987). The ATM price is almost linear in the vol, hence

$$\sigma_{\mathrm{ATM}}(T) \approx \mathbb{E}\big[\sqrt{\bar{v}_T}\,\big] < \sqrt{\mathbb{E}[\bar{v}_T]} = \sqrt{\theta + (v_0 - \theta)\,\tfrac{1-e^{-\kappa T}}{\kappa T}}\,:$$

  the vol of vol lowers the ATM vol, more at long maturities (the dispersion of $\bar{v}_T$ grows), and a negative correlation lowers it further. The ATM term structure depends on all five parameters, not only on v_0, θ, κ.
- *Skew.* At short maturity the ATM skew has a finite limit (Gatheral, 2006, ch. 3),

$$\frac{\partial \sigma_{\mathrm{imp}}}{\partial y}\Big|_{y=0,\ T\to 0} = \frac{\rho\,\xi}{4\sqrt{v_0}},$$

  and once $\kappa T \gg 1$ it decays like $1/T$. The equity skew grows as the maturity shrinks (roughly like $T^{-1/2}$): one parameter set cannot match both ends.
- *Forward smile.* The same ξ and ρ act on every period, so the smile seen from a future date keeps its skew and curvature, with a level that depends on the variance at that date: unlike local vol, the future smile is random and does not flatten.

Closed form. Heston is affine: the characteristic function of $\ln X_T$ is exponential-affine in v_0, $\varphi(u) = \exp(C(T,u) + D(T,u)\,v_0)$, with C, D solutions of Riccati equations. In the form that is continuous in u (Albrecher et al., 2007), with $b = \kappa - \rho\xi i u$, $d = \sqrt{b^2 + \xi^2(iu + u^2)}$, $g = (b-d)/(b+d)$,

$$\varphi(u) = \exp\Big(\frac{\kappa\theta}{\xi^2}\Big[(b-d)T - 2\ln\frac{1-ge^{-dT}}{1-g}\Big] + \frac{b-d}{\xi^2}\,\frac{1-e^{-dT}}{1-ge^{-dT}}\,v_0\Big).$$

European options by the COS method (Fang and Oosterlee, 2008); Oosterlee and Grzelak (2019, sec. 8.4.1): $\ln X_T$ is truncated to $[a,b]$ = mean ± 30 standard deviations (from the cumulants); on it the density is a cosine series whose coefficients are $\mathrm{Re}[\varphi(u_n)e^{-iu_n a}]$, $u_n = n\pi/(b-a)$; the payoff coefficients are closed-form, so the price is a sum of $N = 1024$ terms (exponential convergence). Puts are integrated, calls follow by parity.

Calibration. An optimisation: five parameters cannot reprice all the quotes, so the fit has errors (Oosterlee and Grzelak, 2019, sec. 8.2.3).

1. *Targets*: every node as an undiscounted call per unit forward, $c_{ij} = c_{\mathrm{BS}}(k_{ij}, T_i, \sigma_{ij})$, $k_{ij} = K_j/F(T_i)$ (prices, not implied vols: no inversion inside the search).
2. *Weights*: each maturity divided by its ATM Black vega $\mathcal{V}_i = \sqrt{T_i}\,\phi(\sigma_i^{\mathrm{ATM}}\sqrt{T_i}/2)$, so a price error at the money reads as an implied vol error and all maturities weigh alike; in the wings the residual is the vol error times $\mathcal{V}_{ij}/\mathcal{V}_i$, so the wings count by their vega.
3. *Objective*:

$$\min_{v_0,\kappa,\theta,\xi,\rho} \sum_{i,j} \left(\frac{c^{\mathrm{Heston}}(k_{ij},T_i) - c_{ij}}{\mathcal{V}_i}\right)^2,$$

   with bounds $v_0, \theta \in [10^{-4}, 1]$, $\kappa \in [0.01, 20]$, $\xi \in [0.01, 5]$, $\rho \in [-0.99, 0.99]$; start from the shortest ATM variance for v_0, the average ATM variance for θ, $\kappa = 1.5$, $\xi = 0.5$, $\rho = -0.6$.

4. *Search*: bounded trust-region least squares with a finite-difference Jacobian, on a cheaper COS (256 terms, ±12 standard deviations); then a polish on the full-precision COS and one Newton step on the exact Hessian.
5. *Report*: errors per maturity in vol points (ATM, rms, max), Feller, parameters on a bound.

The errors are largest at the first maturities and in the wings (the short-dated skew). The parameters are poorly identified: κ, θ and ξ trade off (fits of almost the same quality along a curve), so they move between calibrations while the fitted surface barely does. In practice: a global search or yesterday's parameters as start, some parameters fixed, a penalty on parameter changes, the calibration restricted to the product's maturities, or richer models (time-dependent parameters, Andersen and Piterbarg, 2010; jumps, Bates, 1996).

Simulation. The quadratic-exponential scheme of Andersen (2008) (see also Oosterlee and Grzelak, 2019, sec. 9.3.4). Given v_t, the next variance has mean $m = \theta + (v_t - \theta)e^{-\kappa\Delta}$ and variance $s^2 = v_t\,\xi^2 e^{-\kappa\Delta}(1 - e^{-\kappa\Delta})/\kappa + \theta\,\xi^2(1 - e^{-\kappa\Delta})^2/(2\kappa)$; with $\psi = s^2/m^2$, it is drawn as $a(b + Z)^2$ when $\psi \le 1.5$ (moments matched) and from a mass at zero plus an exponential tail otherwise, which keeps it non-negative without truncation. $\ln X$ is then stepped with the time integral of v approximated by the average of the two ends and the drift chosen so that X stays an exact martingale. Two normals per step. The scheme is accurate with large steps (quarterly steps reprice the closed form within the Monte Carlo error, Feller violated).

Sensitivities. The parameters depend on the quotes through the optimum, so each sensitivity runs through the calibration:

$$\text{quotes} \;\to\; (F(T_i),\; c_{ij}) \;\to\; p^* = (v_0, \kappa, \theta, \xi, \rho) \;\to\; \text{price (closed form or paths)}.$$

The optimum satisfies $\nabla_p L(p^*, m) = 0$ for every market input m; differentiating this identity (implicit function theorem),

$$\frac{\mathrm{d}p^*}{\mathrm{d}m} = -H^{-1}\,\frac{\partial^2 L}{\partial p\,\partial m}, \qquad H = \nabla_p^2 L(p^*, m),$$

the derivative of a re-optimisation without re-optimising (parameters on a bound held fixed).

- *Vega per node*: every parameter depends on many nodes (v_0 on the short ATM vols, κ and θ on the long maturities with opposite signs, ρ on the skew, ξ on the curvature), so a price has vega on the whole surface, with signs, also far from its own strike and maturity. The total vega is close to the Black–Scholes one, the buckets are not.
- *Delta*: a higher spot raises the forwards (the paths scale with them) and moves every node to a lower moneyness, which changes the targets and hence the parameters; both effects go through the chain above.
- *Rates, repo, dividends*: through the forwards, as the delta without the move of the spot.

Closed-form and Monte Carlo (pathwise) greeks agree within the Monte Carlo error.

Limits.

- *Fit*: the short-dated skew and the term structure of the skew are out of reach of five constant parameters; vanillas and near-expiry exotics are mispriced in the short wings.
- *Identifiability*: unstable parameters and vega buckets from one calibration to the next.
- *Dynamics tied to today's smile*: the vol of vol and correlation that fit today's curvature and skew also set the forward smile; there is no separate control.

- *Feller usually violated*: a mass of the variance near zero; the simulation needs a scheme built for it.
- *Constant correlation and long-run level*: no regimes.
- *Local optimisation*: the result can depend on the start and on the weights.

Heston is the reference for the dynamics of the vol and has a fast closed form, but does not reprice the surface; local vol does the opposite, and the stochastic local vol model of §2.5 combines the two.

2.5 Heston stochastic local volatility

Dynamics. The Heston variance multiplied by a leverage function of time and spot:

$$\frac{\mathrm{d}X_t}{X_t} = L(t, X_t)\,\sqrt{v_t}\;\mathrm{d}W_t^1, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\;\mathrm{d}t + \eta\,\xi\sqrt{v_t}\;\mathrm{d}W_t^2, \qquad \mathrm{d}\langle W^1, W^2\rangle_t = \rho\;\mathrm{d}t,$$

$(v_0, \kappa, \theta, \xi, \rho)$ the calibrated Heston parameters of §2.4, the vol of vol scaled by a mixing factor $\eta \in (0, 1]$. The instantaneous variance is $L^2 v_t$: the stochastic variance gives the dynamics (forward smile, vol of vol), the leverage corrects its level spot by spot so that the marginals are those of local vol (van der Stoep et al., 2014; Oosterlee and Grzelak, 2019, sec. 10.2).

The leverage. By Gyöngy's theorem (Gyöngy, 1986) a process with stochastic instantaneous variance has the marginals of the local vol model whose local variance is the expected instantaneous variance given the spot. The SLV model has the market marginals if $L^2(t,x)\,\mathbb{E}[v_t \mid X_t = x] = \sigma_{\mathrm{loc}}^2(t,x)$, hence

$$L^2(t,x) = \frac{\sigma_{\mathrm{loc}}^2(t,x)}{\mathbb{E}[v_t \mid X_t = x]},$$

σ_{loc} the local vol of §2.3. Where the Heston variance on the paths through (t,x) is too high for the market, $L < 1$, where too low, $L > 1$. Limits: on a Heston market $L = 1$; for $\eta \to 0$ the variance is deterministic and the model is local vol. Every η reprices the same vanillas: η only moves the dynamics, from local vol ($\eta \to 0$, flat forward smile) towards Heston ($\eta = 1$).

Calibration. No optimisation beyond its two components:

1. the local vol from the surface (§2.3), exact;
2. the Heston parameters from the surface (§2.4), with $\xi \to \eta\,\xi$;
3. the conditional expectation $\mathbb{E}[v_t \mid X_t]$, on a dedicated simulation. It depends on the law of (X_t, v_t), hence on the leverage at earlier times (a McKean SDE): it is built forward in time on particles (particle method, Guyon and Henry-Labordère, 2012) and stored on a grid (50 time rows a year, 41 nodes per row). At each row:
   (a) 41 nodes span the 0.1%–99.9% quantiles of $\ln X_t$ over the particles, widened by 1% of the range (no particle exactly on an end node);
   (b) $\mathbb{E}[v_t \mid \ln X_t]$ at the nodes is the average of v_t with tent (piecewise linear) weights, each particle splitting its weight between its two nodes; nodes with few particles are shrunk to the overall mean of v_t (a pseudo-count of 5% of the particles per node), and the result is floored at 10^{-4};
   (c) the smoother averages over a node spacing h, a bias of order h^2 (about -0.6 vol points at 3Y ATM on the test surface): the estimate used is the Richardson combination $(4\,\mathbb{E}^h - \mathbb{E}^{2h})/3$ of the estimates on the nodes and on every other node, which removes the leading term (-0.2);

(d) each particle gets $L = \sigma_{\text{loc}}/\sqrt{\mathbb{E}[v \mid X]}$ at its own $\ln X_t$ (capped at 10), and the step is taken.

Two speeds: the Heston parameters now and then, this table whenever the surface moves (Heston held);

4. the mixing factor η is not determined by the vanillas: it is set to reprice products that depend on the dynamics (forward-start options, barriers, cliquets) when quotes exist, or by judgement.

The leverage is far from 1 where Heston and the market disagree: at short maturities it supplies the steep short-dated skew and wings that Heston misses; at long maturities it removes the excess curvature of the Heston smile.

Simulation. QE step for v (§2.4), log step of X with L frozen over the step (the Heston log-step coefficients scaled per path by L), exact martingale correction per path. The pricing paths read the stored $\mathbb{E}[v \mid X]$ at their own $\ln X_t$ (linear on the nodes, linear in time between rows); σ_{loc} and F come from the market.

Sensitivities. By convention the Heston parameters are held; the leverage moves with the market through the local vol (vol nodes), the forward (spot, repo, dividends, curve) and the table $\mathbb{E}[v \mid X]$, recalibrated on the moved market:

$$\text{quotes} \;\to\; \sigma_{\text{loc}}(t,y),\ F(t),\ \mathbb{E}[v \mid X] \;\to\; L = \sigma_{\text{loc}}/\sqrt{\mathbb{E}[v \mid X]} \;\to\; X_t \;\to\; S_t = F(t)\,X_t \;\to\; \text{payoff}.$$

Following Savine (2018, ch. 13): the pricing gives the direct sensitivities and $\partial V/\partial \mathbb{E}$ (including the move of each path on the table), and the market sensitivity is $\mathrm{d}V/\mathrm{d}m = \partial V/\partial m + \sum \partial V/\partial \mathbb{E} \cdot \mathrm{d}\mathbb{E}/\mathrm{d}m$, the second term by the adjoint of the table's calibration (one backward pass through it; for a portfolio on one underlying the trades' $\partial V/\partial \mathbb{E}$ summed, one pass). Both terms are needed: holding the table fixed misplaces the vega of a vanilla between the strikes of its expiry and, at long maturities, misses part of it (a quarter of the vega of a 5Y put on the test surface). The adjoint requires a smooth estimator (few nodes, end nodes off the particles): its exact derivative follows the particle noise of each node and grows along the calibration, and with 100 nodes, or with the Richardson-corrected estimate, it diverges. Hence smoothing for risk (Savine, 2018): two tables on the same nodes and particles, the corrected one for the price and the 41-node one for the sensitivities (the paths take the value of the first and the derivative of the second). Against bump-and-recalibrate of the corrected model the sensitivities agree within 3% on the node of a vanilla and on the parallel vega, and up to 0.05 per vol point on other nodes. In exact SLV a vanilla does not depend on the Heston parameters (the leverage compensates them), so the convention loses nothing for vanillas: their delta and vega per node are close to the local vol ones. For an exotic the dynamics is frozen; its sensitivity to the dynamics (to η, to a Heston recalibration) is a separate model-risk number.

Limits.

- *Dynamics by choice*: η is not identified by the vanillas, and exotic prices depend on it.
- *Estimator*: Monte Carlo noise on the vanillas (more paths than local vol), and choices (nodes, kernel, shrinkage, floor, cap) that matter in the wings. On a steep equity surface the short-dated downside is not repriced exactly (a few tenths of a vol point, systematic: it does not vanish with more particles).
- *Cost*: a Heston calibration now and then, the two tables whenever the surface moves, two path simulations per price, and the adjoint of the calibration for the market sensitivities.

- *Sensitivities of a smoother model*: the greeks are those of the uncorrected estimator, not exactly of the model that prices. A deterministic estimate of $\mathbb{E}[v \mid X]$ (the Fokker–Planck PDE of the joint density) would make the calibration smooth and remove the trade-off.
- *Greeks by convention*: the dynamics risk is not in the greeks.
- *Inherited*: the interpolation dependence of local vol and Feller violated in the Heston part.

SLV is the standard equity model for path-dependent exotics: it reprices the hedging vanillas and leaves the dynamics as an explicit choice.

2.6 Gaussian short-rate models

Dynamics. The short rate is a sum of Gaussian factors plus a deterministic shift:

$$r(t) = \sum_i x_i(t) + \varphi(t), \qquad \mathrm{d}x_i = -\kappa_i\, x_i\, \mathrm{d}t + \sigma_i(t)\, \mathrm{d}W_t^i, \qquad x_i(0) = 0,$$

with correlated Brownians; one factor is Hull–White (Hull and White, 1990) (G1++), two factors G2++ (Brigo and Mercurio, 2006, ch. 3–4). The vols are constant or piecewise constant in time. The shift φ is chosen so that the model reproduces the market discount curve exactly, so only the vol parameters are calibrated. With $V(t,T) = \mathrm{Var}\big(\int_t^T \sum_i x_i\big)$ (exact integrals over the vol pieces) the zero-coupon bonds are

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(\tfrac{1}{2}\big[V(t,T) - V(0,T) + V(0,t)\big] - \sum_i B_i(T-t)\, x_i(t)\Big), \quad B_i(\tau) = \frac{1 - e^{-\kappa_i \tau}}{\kappa_i}.$$

Closed forms.

- *Caplets*: an option to exchange two zero-coupon bonds, whose log ratio is Gaussian (Margrabe, 1978); the index curve enters through a deterministic multiplicative basis to the discount curve.
- *Swaptions*: either the Gaussian swap-rate approximation (the swap rate linearised in the factors around zero, normal vol from its variance), or exact: the swap as a portfolio of bonds under the forward measure of the expiry, by Jamshidian (1989) with one factor, or with two by Brigo and Mercurio (2006) (Gauss–Hermite on the first factor, closed form on the second).

Calibration. Least squares on caplets and / or swaptions: ATM (or given strikes) caplets up to a term, from the normal caplet vols, and swaptions from the cube. Each residual is the model minus market price divided by the normal vega, close to a normal vol error. Parameters inside bounds; a parameter can be held fixed. Hull–White with a piecewise constant $\sigma(t)$ on the caplet fixings matches the ATM caplets exactly, with the mean reversion held (caplets alone do not determine it). With two factors, swaptions of several tenors are needed to pin the correlation, which caplets leave undetermined.

Simulation. Exact joint Gaussian transition of the factors and of their time integrals; the bank-account discount is $D(0,t) = P(0,t)\, e^{-\int_0^t \sum_i x_i - V(0,t)/2}$. Swap exposures (mark-to-market, expected positive and negative exposure, potential future exposure) are computed on these paths.

3 Products

Each product with its payoff and how it is priced: in closed form on the market data of §1, or by Monte Carlo on a model of §2. Sensitivities are derivatives of the price w.r.t. the market quotes: delta and gamma on the spot, vega per vol node, rho per curve pillar, sensitivities to the dividend amounts; theta is the derivative w.r.t. the time to maturity with rates and dividend inputs held.

3.1 Linear rate products

Notation: P_d the discount (overnight) curve of the currency, P_f the projection curve of the index, N the notional, τ the accrual fraction.

- *Deposit*: NPV $= N\,P_d(0,s) - N(1+r\tau)\,P_d(0,e)$.
- *FRA*: NPV $= \pm N\tau\,(F-K)\,P_d(0,T_1)/(1+F\tau)$, F the forward of the index on P_f, settled at the start; zero once started.
- *Fixed leg*: $\sum_i NK\tau_i\,P_d(0,p_i)$.
- *Floating leg* (Ibor): coupon $N(g\,F_i+m)\,\tau_i$ discounted at $P_d(0,p_i)$, gearing g, spread m, forward $F_i = \big(P_f(0,s_i)/P_f(0,e_i)-1\big)/\tau_i^{\text{idx}}$.
- *Swap*: payer = floating − fixed leg; par rate = floating leg/annuity.
- *OIS*: compounded overnight coupon $\big(P_d(0,s_i)/P_d(0,e_i)-1\big)/\tau_i$ on the discount curve itself, so the floating leg telescopes to $N\big(P_d(0,s)-P_d(0,e)\big)$.

3.2 Caps, floors and collars

A cap (floor) is a strip of calls (puts) on the Ibor fixings, the first period excluded:

$$\text{caplet}_i = N\tau_i\,P_d(0,p_i)\,\text{Opt}(F_i,K,\sigma_i,T_i^{\text{fix}}),$$

Opt the formula of Bachelier (1900) on normal caplet vols, of Black (1976) on lognormal ones, σ_i the caplet vol at (T_i^{fix},K); a caplet already fixed is the intrinsic value on the fixing, a paid one is dropped. Collar = cap − floor. Under a Gaussian short-rate model the caplets have their own closed form (§2.6), used in its calibration.

3.3 Swaptions

European, physically settled, into a fixed-vs-Ibor swap. With the annuity of the fixed leg $A = \sum_i N\tau_i\,P_d(0,p_i)$ and the forward swap rate $F = \text{PV}_{\text{float}}/A$,

$$V = A\,\text{Opt}(F,K,\sigma,T), \qquad \sigma = \sigma_{\text{cube}}(T,\text{tenor},K-F),$$

Bachelier (normal cube) or Black-76 (lognormal); see Brigo and Mercurio (2006). Model prices under G1++ / G2++: §2.6.

3.4 Vanilla options

Calls and puts on an equity, $\phi = \pm 1$, payoff $(\phi(S_T-K))^+$; European or American.

Black–Scholes (Black and Scholes, 1973; Merton, 1973). European on the forward F of §1.3 and the surface vol at (K, T):

$$V = P(0,T)\,\phi\,\big(FN(\phi d_1) - KN(\phi d_2)\big), \qquad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

Two dividend models (Haug et al., 2003): *discrete*, on the escrowed spot $S_0 - \mathrm{PV}_{\mathrm{div}}(T)$ with carry $b = r - q_r$; *continuous*, with the dividends replaced by the equivalent yield $q_{\mathrm{eff}}(T)$, held fixed when the spot moves (sticky yield), carry $b = r - q_{\mathrm{eff}} - q_r$. The two give the same price and different deltas. With only a yield in the data, $b = r - q - q_r$.

American (Bjerksund and Stensland, 1993), see also Haug (2007). A call with carry b is exercised at a flat boundary I:

$$I = B_0 + (B_\infty - B_0)(1 - e^{h(T)}), \qquad h(T) = -(bT + 2\sigma\sqrt{T})\frac{B_0}{B_\infty - B_0},$$

$$B_\infty = \frac{\beta}{\beta - 1}K, \qquad B_0 = \max\Big(K, \frac{r}{r-b}K\Big), \qquad \beta = \tfrac{1}{2} - \frac{b}{\sigma^2} + \sqrt{\Big(\frac{b}{\sigma^2} - \tfrac{1}{2}\Big)^2 + \frac{2r}{\sigma^2}},$$

value $\alpha S^\beta - \alpha\,\phi(S,T,\beta,I,I) + \phi(S,T,1,I,I) - \phi(S,T,1,K,I) - K\phi(S,T,0,I,I) + K\phi(S,T,0,K,I)$ with $\alpha = (I-K)I^{-\beta}$; $S-K$ above I; the European value when $b \geq r$. Puts by the transformation $P(S,K,r,b) = C(K,S,r-b,-b)$.

Monte Carlo (European). Paths of one of the models of §2, calibrated to the market, on a uniform time grid ending at the expiry; price $P(0,T)$ times the mean payoff, with its standard error. The normals are drawn from a fixed seed, so a repricing with bumped market data uses the same random numbers.

3.5 Digital options

European, paid at T on $\phi(S_T - K) > 0$: cash-or-nothing (amount A) or asset-or-nothing (S_T). *Black–Scholes*, vol at the strike: $D^{\mathrm{cash}} = A\,P(0,T)\,N(\phi d_2)$, $D^{\mathrm{asset}} = P(0,T)\,F\,N(\phi d_1)$. This ignores the smile slope. *On the smile* the unit cash digital is $-\phi\,\partial V/\partial K$ of the vanilla priced at its own vol, $P(0,T)\,N(\phi d_2) - \phi\,\mathcal{V}\,\partial\sigma/\partial K$ ($\mathcal{V}$ the vanilla vega), computed as the centred spread $\phi\,[V(K - hK) - V(K + hK)]/(2hK)$; asset-or-nothing $= K\,D^{\mathrm{cash}}_{A=1} + \phi\,V(K)$. With a negative skew the smile lowers the digital put and raises the digital call; the gap is first order (one vega times the skew). *Monte Carlo*: the indicator has no pathwise derivative, so it is replaced by the sigmoid $1/(1 + e^{-\phi(S_T-K)/(\varepsilon K)})$ of width ε (1% of the strike by default; $\varepsilon = 0$ is the exact indicator, with no pathwise delta or vega).

3.6 Barrier options

European call or put, knocked in or out when the spot reaches the barrier H from above (down) or from below (up); the barrier is monitored continuously or on given dates (a single date at expiry is the barrier observed at expiry only); no rebate. Knock-out = vanilla − knock-in.

Black–Scholes (one vol: the surface vol at (K, T)):

- continuous monitoring: the closed form of Reiner and Rubinstein (1991) with carry b (Haug, 2007);

- at expiry only: exact by vanillas and cash digitals struck at H, e.g. the down-and-in put is $\mathrm{Put}(H) + (K - H)\,\mathrm{DigitalPut}(H)$ for $K > H$;
- n dates: the continuous formula at the shifted barrier $H\,e^{\pm 0.5826\,\sigma\sqrt{T/n}}$ of Broadie et al. (1997) (moved away from the spot).

Monte Carlo. Continuous monitoring by the Brownian-bridge probability of not crossing the barrier between two simulated points (Beaglehole et al., 1997; Glasserman, 2003), multiplied over the steps:

$$\prod_i \left(1 - e^{-2\,\ln(S_i/H)^+\,\ln(S_{i+1}/H)^+/(\sigma_H^2\,\Delta t_i)}\right) \quad \text{(down; } \ln(H/S) \text{ for up)},$$

σ_H the surface vol at (H, T): exact for a lognormal model with that vol, and continuous in the path (pathwise sensitivities). Discrete monitoring checks the barrier on the dates, the indicator smoothed as for the digitals.

3.7 Autocallable

A strike K (the barriers in % of it); coupon fixings t_i with rates c_i, barriers B_i^c and payment dates p_i, with or without memory; autocall fixings u_j with barriers B_j^a; at the last coupon fixing a put knocked in below B^p (observed at that date only), with participation π. Valued on the investor's side, coupons minus put, per unit notional; on each path, with $\alpha = 1$ until the first autocall and m the coupons unpaid so far:

$$V = N\,\mathbb{E}\Big[\sum_i \alpha_{t_i}\,\mathbb{1}\{S_{t_i} \geq B_i^c\}\,c_i(1 + m_{t_i})\,P(0,p_i) \;-\; \alpha_T\,\mathbb{1}\{S_T < B^p\}\,\pi\,\frac{(K - S_T)^+}{K}\,P(0,p_n)\Big],$$

m reset when a coupon is paid and increased by one when it is missed ($m \equiv 0$ without memory); α set to 0 when $S_{u_j} \geq B_j^a$. Monte Carlo on a model of §2 (local vol by default), on the fixing dates merged with a weekly grid. Each indicator is a sigmoid of width εK so that delta and vega exist pathwise. With the autocall out of reach and no put, the product is a strip of digitals; with no coupons, a put knocked in at expiry (§3.5, §3.6).

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